

Article Generation Pipeline

Prompt builder

Retrieved passages become ranked context blocks with strict output instructions: **TITLE:** and **ARTICLE:**

Local LLM generation

OpenAI-compatible endpoint, controlled temperature, max-token budget, one repair attempt

Evaluation

ROUGE, BERTScore, title similarity, section-count difference, article-length ratio

Database upsert

generated text, reference text, and metrics are stored in Postgres

Validation layer

parse title/body, reject malformed output, repair once if needed

Runner behavior

- ▶ single-article or batch execution
- ▶ schema creation for **generated_articles**
- ▶ skip-existing logic from database state
- ▶ per-article error isolation for long runs

Cost profile

Generation dominates runtime. Retrieval, prompt assembly, and database writes are much cheaper, while larger k and larger token budgets mainly increase latency.

Stored output

Each run persists the generated article together with the withheld reference text and evaluation metrics, enabling direct comparison across prompts, models, and retrieval settings. With more project time, this can be used for improving the prompts/structure.

Conclusion & Future Work

Conclusion

The implemented system already provides a complete retrieval-to-generation baseline: it loads article metadata and reference text from Postgres, retrieves evidence from FAISS-indexed passages, builds a constrained prompt, generates a Wikipedia-style article draft with a local LLM, evaluates the result against the withheld reference article, and stores outputs plus metrics for reproducible comparison.

What already works well

- ▶ end-to-end reproducible pipeline
- ▶ explicit grounding in retrieved passages
- ▶ lexical, semantic, and structural evaluation
- ▶ resume-friendly batch processing via upserts

Future work

- ▶ section-wise RAG instead of single-shot generation
- ▶ better topic inference and query expansion
- ▶ stronger redundancy reduction before prompting
- ▶ tuning of k , prompt length, and model choice
- ▶ richer factuality checks beyond overlap metrics

Takeaway

The strongest next gains will likely come from better retrieval structure and more controlled generation rather than from rebuilding the pipeline itself.